

# Hyder Mohyuddin

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**portfolio** → [hydermohyuddin.com](https://hydermohyuddin.com) – 15+ projects, write-ups & live demos – my work in full  
**profiles** → [github.com/hmohyud](https://github.com/hmohyud) · [linkedin.com/in/hyder-mohyuddin](https://linkedin.com/in/hyder-mohyuddin)

## // EDUCATION

**B.S. Computer Science**, Minor in Visual Arts – University of Chicago

Mar 2025

*Coursework: Data Visualization, Theory of Algorithms, Discrete Math, Systems Programming II, Complexity Theory, Data Science (Pandas & Statistics), Art & Machine Intelligence, Interactive Devices*

## // SKILLS

|                    |                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Languages</b>   | Python, JavaScript, TypeScript, Java, C, C#, Swift, SQL, Assembly                                                |
| <b>Web</b>         | Full-stack development: React, React Native, HTML5, CSS, D3.js, REST APIs, WebSockets, Git, Linux, Figma         |
| <b>AI / ML</b>     | PyTorch, machine learning & generative AI (Stable Diffusion, LLMs), computer vision (OpenCV), prompt engineering |
| <b>Fabrication</b> | Autodesk Fusion, 3D printing, Arduino & electronics                                                              |

## // WORK EXPERIENCE

**AI Software Engineer – Stable Diffusion Research**, *University of Chicago*

Dec 2023 – May 2025

- Built SPIM (Salavon's Pathology Inducing Machine), a full-stack research platform that hot-edits diffusion-model tensors mid-generation to study what each layer contributes – 800+ commits (~96% of the codebase) in Python and React, from browser UI to GPU servers.
- Architected the serving stack: a secure auth gatekeeper farming generation jobs to standby GPU render servers (machines I built and tuned myself), with custom memory management and code obfuscation against reverse engineering.
- Enabled pattern-based layer targeting (e.g. `diffusion_model.*.1.norm`), letting a single weight/bias formula rewrite dozens of tensors across the U-Net or text encoder in one experiment.
- Built a randomized-formula engine (weighted-probability generation with automatic simplification) plus statistical-novelty filters and logging to quantify what large experiment runs produce.
- Ran AnimateDiff, ControlNet, and LoRA simultaneously for animations; assembled ultra-high-resolution stills via Voronoi-based tiling, beyond standard software limits.
- Cut GPU costs by gating full-resolution renders behind cheap low-res previews that must pass a statistical quality check.
- Became the production tool of Jason Salavon's studio – used to create his 2024 solo show "History Painting" (TAI Modern, Santa Fe) – and demoed to investors, helping secure funding.

**Full-Stack Developer** (*Contract*)

2024 – Present

- Revived Lavoët, a Parisian trunk-maker founded 1869, as a working digital brand: an interactive catalogue raisonné in inline SVG, a parametric pattern studio, and its c.1930 monogram rebuilt via computer vision – delivered as zero-dependency single-file apps.

- Built an aerospace part-sourcing tool: concurrent async scraping (Playwright + static fallbacks) with deduplication, and a local llama.cpp model returning validated structured JSON.
- Developed AutoLens, a vehicle-identification app (Expo/React Native) with on-device image processing and server-side model inference.

### **Web Developer** (*Contract*)

2024 – Present

- Built and run the Compassion Course platform (React + Firebase) serving 50,000+ participants in 120+ countries – scheduled lessons, a custom CMS, and a team dashboard with Kanban boards and shared whiteboards.
- Designed and shipped production sites for clients: brand landing pages (Glo Pop, Gorilla Greens), an art-therapy practice, an interfaith charity with a nomination pipeline, and a poetry collection – largely framework-free HTML/CSS/JS.

### **Software Project Lead** – *UChicago Booth School of Business*

Dec 2023 – Nov 2024

- Visualized 200GB of environmental data on an interactive Three.js globe with dynamic Leaflet palettes, fed by Python pipelines that automated processing of compiled SQL data.
- Led the team's weekly Agile-style cycles, aligning priorities and ownership.

### **Tutor & Youth Coach** – *Self-employed & NGOs, Chicago*

2016 – Present

- 1,000+ hours of individualized instruction to 50+ students in math, physics, chemistry, and writing.
- Code Platoon: taught evening/weekend coding bootcamp to 30 veterans and active-duty service members.
- Code Your Dreams: taught a 7-week intro coding course to 40 inner-city high-school students; trained students in Youth Initiated Mentoring.
- MetroSquash (2012–2016, volunteer): weekly after-school squash coaching for inner-city students.

### **Technology Support Assistant** – *UChicago Humanities Computing*

2018 – 2021

- Imaged, secured, and maintained 100+ Humanities Division computers; automated OS/app deployment and supported faculty and staff.

### // SELECTED PROJECTS

- **Pocket Sous Chef** – iOS app on the App Store: pantry photos to structured recipes via an AI text-and-vision pipeline (SwiftUI + SwiftData).
- **Typometry** – client-side typing-analytics web app: keyboard heatmaps, bigram timing, behavioral profiling (React + Vite).
- *Every project and client build above has a full write-up and live demo at [hydermohyuddin.com/projects](https://hydermohyuddin.com/projects).*

### // INTERESTS

**Languages** – English & Gujarati (fluent), Hindi & French (intermediate), Arabic & Japanese (beginner) · **Arts** – digital art, pencil sketching, stencil design, 3D design & printing, painting · **Activities** – squash, skiing, biking, chess, video games